

Enterprise Knowledge Base Platform (KBS)

A collaborative, governed Single Source of Truth for organizational knowledge

Create & Collaborate

Discover & Traverse

Govern & Preserve

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Executive Summary & Product Vision

What KBS is, why it exists, and the value it delivers to the enterprise.

Product Overview

The **Enterprise Knowledge Base Platform (KBS)** is a centralized, collaborative **Single Source of Truth (SSOT)** for modern, multi-team organizations. It gives every department a unified workspace to author, discover, interconnect, and govern technical specifications, product requirements, engineering runbooks, operational policies, and organizational decisions. KBS combines real-time collaborative authoring with deep knowledge-graph interconnectivity, enterprise-grade dynamic access control, and native developer-ecosystem integrations (GitHub, GitLab, Jira).

Core Value Proposition

Single Source of Truth One searchable repository replaces scattered wikis, drives, and chat threads.	Frictionless Collaboration Distributed teams co-author, review, and comment simultaneously with zero lockouts or lost work.	Impact Safety Bidirectional backlinks and visual mapping warn downstream owners before breaking changes.
Developer Sync GitHub/GitLab/Jira integration keeps specs, code and sprint boards aligned.	Flexible Governance Hierarchical permissions with explicit single-document overrides — no over-provisioning.	Auditability Full snapshot versioning, 1-click rollback, and an immutable change ledger.

Operating Model — Three Pillars

Every capability in KBS rolls up into three pillars that together produce the authoritative enterprise SSOT:

 Create & Collaborate Author, co-edit and discuss knowledge in real time. (BF-01 – BF-03)	 Discover & Traverse Search, connect and understand impact before change. (BF-04 – BF-06)	 Govern & Preserve Secure, publish, version and audit knowledge safely. (BF-07 – BF-10)
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This BRD consolidates five source specifications (Overview, Users & Roles, Core Capabilities, Access Control, and Business Workflows) into a single, final, implementation-ready reference.

Problem Statement

As organizations scale, knowledge generation outpaces the organization's ability to index, maintain and connect it — creating seven recurring bottlenecks.

1. Fragmentation & Siloing Rigid folders trap documents in one department; cross-functional discovery fails and duplicate work follows.	2. Edit Collisions Lock-based or "last write wins" tools force manual reconciliation, discouraging live documentation.	3. Binary Access Dilemma Admins either over-provision access to a whole workspace, or force unmanaged workarounds — both are risky.
4. Dependency Blindspots One-way links hide what depends on a document, so edits silently break downstream runbooks.	5. Documentation Drift Code evolves in Git while wikis stay static, causing dangerous gaps during incidents.	6. Unstandardized Docs Specs, ADRs and postmortems use arbitrary formats; critical sections get omitted.
7. Loss of Version Integrity & Compliance Blindspots Without named snapshots, diffing, and immutable audit trails, accidental changes can't be confidently reverted and compliance teams cannot reconstruct who changed what, when.		
Net effect: organizational inefficiency, rework, security risk, and operational outages — all traced back to knowledge that is scattered, unsafe to share, or silently out of date.		

Business Goals & Product Scope

Seven business goals directly answer the seven problems above; scope defines the Phase 1 delivery boundary.

Primary Business Goals

Ref	Business Goal	Strategic Purpose
BG-01	Consolidate Organizational Knowledge	One enterprise-wide repository replacing ad-hoc wikis, drives and local files.
BG-02	Accelerate Cross-Team Collaboration	Real-time simultaneous authoring and discussion so teams finalize specs faster.
BG-03	Enforce Secure, Flexible Governance	Protect sensitive data while enabling safe single-document sharing.
BG-04	Prevent Knowledge Rot & Broken Dependencies	Automated backlinks and pre-change impact warnings.
BG-05	Bridge Engineering & Business Knowledge	Sync KBS with GitHub/Jira so specs track active code and sprints.
BG-06	Standardize Documentation Excellence	Curated and custom blueprint libraries with approval gates.
BG-07	Ensure Full Compliance & Traceability	Non-destructive history, instant restore, immutable event auditing.

Scope — Phase 1 Baseline

✓ In Scope

- 4-tier hierarchy: Organization → Department → Project → Document → Sub-Document
- Real-time collaborative Markdown authoring, live presence, conflict-free sync
- Curated & custom blueprint libraries with scoping and approval
- Mermaid diagramming and freehand visual canvas
- GitHub PR/Issue cards, live code snippets, Docs-as-Code sync
- Dynamic Access Control with inheritance, overrides, live demotion
- Backlinks, interactive knowledge graph, pre-change impact warnings
- Version history, visual diff, non-destructive restore
- Move/rename resilience, Word/Markdown import, PDF/HTML export
- Command palette search with zero-leak security & deep-linking
- Enterprise audit logging for security, permissions and lifecycle

✗ Out of Scope (Deferred)

- Public / anonymous internet publishing (internal, authenticated platform only)
- Third-party desktop file-sync agents (e.g. Dropbox/Drive file stream)
- Automated natural-language translation of documents
- Native video/audio hosting and transcoding (files may still be attached)

Key Assumptions

- Organization provides an Identity Provider for auth & department membership
- GitHub integrations use authorized OAuth / App credentials
- Teams want transparent discovery by default, with security applied to sensitive assets

Users, Roles & Collaboration Model

Container stewards manage structure; document roles scale collaboration on individual pages. Every role can be granted to an individual or to a reusable User Group.

Container Administrative Roles

Role	Scope	Core Responsibilities
System Admin	Organization-wide	Global governance, user management, integrations, org taxonomy, global audit, trash recovery.
Department Head	Department	Creates/archives Projects, appoints Project Leads, sets department default access, approves department blueprints.
Project Lead	Project	Manages project membership & metadata, creates project blueprints, configures GitHub links, organizes documents.

Document Collaboration Roles

Role	Core Responsibilities
Co-Owner (Author)	Full editing, access administration, lifecycle control (rename, move, deprecate, delete), publish milestones, rollback, save as blueprint. Multiple co-owners supported per document.
Reviewer	Reviews visual diffs, leaves structured change requests, formally approves & signs off publication.
Editor	Co-authors content, diagrams, checklists, GitHub embeds; cannot delete, move, transfer ownership, or manage sharing.
Commenter	Reads content, opens/replies to comment threads, resolves threads; cannot edit body text.
Viewer	Read-only: content, metadata, graph, exports; read-only comments; cannot author or manage.

User Groups & Cross-Functional Sharing

Named reusable teams (e.g. *@security-team*, *@legal-counsel*) can be assigned any standard role on a Department, Project or Document, with automatic access updates as membership changes. This enables **cross-department document sharing** — e.g. an Engineering-owned document can grant the Legal team *Commenter* access without exposing the rest of Engineering's private projects.

Key Personas

Persona	Primary Role	Motivation
Elena — Head of Engineering	Department Head	Oversight without editing every spec; approves department blueprints.
Marcus — Project Manager	Project Lead	Organizes roadmaps, links Jira/GitHub, invites cross-functional teams.
Aarav & Sarah — Tech Leads	Document Co-Owners	Co-author specs, manage reviewers, publish official versions.
Devin — Software Engineer	Editor	Contributes implementation detail, embeds diagrams, updates checklists.
Priya — Legal / Compliance	Reviewer / Commenter	Reviews privacy sections, signs off on compliance specs.
Liam — New Hire Developer	Viewer	Browses runbooks, traverses the graph to learn the architecture.

Role Capability Matrix

Authoritative entitlement grid across the eight platform roles.

Capability	Admin	Dept Head	Proj Lead	Co-Owner	Reviewer	Editor	Commenter	Viewer
Manage global settings / departments / user groups	✓	✗	✗	✗	✗	✗	✗	✗
Create Projects / manage project settings	✓	✓	✓ (own)	✗	✗	✗	✗	✗
Read content, explore graph & backlinks, export	✓	✓	✓	✓	✓	✓	✓	✓
Real-time co-author & edit document text	✓	✓	✓	✓	✗	✓	✗	✗
Embed diagrams / GitHub snippets / toggle checklists	✓	✓	✓	✓	✗	✓	✗	✗
Add / reply to inline comments & resolve threads	✓	✓	✓	✓	✓	✓	✓	✗ (read-only)
Submit for formal review	✓	✓	✓	✓	✗	✓	✗	✗
Approve / sign-off formal review	✓	✓	✓	✓	✓	✗	✗	✗
Direct publish milestone / rollback to snapshot	✓	✓	✓	✓	✗	✗	✗	✗
Move document / manage sharing & permissions	✓	✓	✓	✓	✗	✗	✗	✗
Soft-delete / restore document / inspect audit logs	✓	✓	✓	✓	✗	✗	✗	✗

Resolution rule: Individual Explicit Grant > Team Group Grant > Project Grant > Department Baseline > Organization Default.
See Section 7 for the full Dynamic Access Control model.

Core Business Capabilities

Eleven foundational capabilities underpin the ten business workflows detailed in Section 8.

ID	Capability	Business Outcome
CAP-01	Rich Authoring, Diagramming & Canvas	Structured Markdown, Mermaid diagrams, freehand canvas, interactive SOP checklists.
CAP-02	Blueprint Libraries & Template Governance	Curated (ADR/PRD/SOP/RCA) and custom blueprints with scoped approval; safe forking.
CAP-03	Real-Time Concurrent Collaboration	Conflict-free co-authoring, live presence/cursors, offline resilience.
CAP-04	Contextual Inline Commenting & Notifications	Text-anchored threads, @user/@team mentions, resolve/reopen lifecycle.
CAP-05	Knowledge Topology & Graph Traversal	Bidirectional backlinks, visual graph, multi-hop pathway discovery.
CAP-06	Developer & GitHub Ecosystem Integration	Live PR/Issue cards, synced code snippets, Docs-as-Code sync, PR-merge alerts.
CAP-07	Portability, Hierarchy & Tree Management	Nested sub-pages, cross-project moves, rename auto-sync, import/export.
CAP-08	Lifecycle, Sign-Off & Non-Destructive Versioning	Draft vs. published states, agile or formal review, visual diff, 1-click rollback.
CAP-09	Unified Permission-Aware Search	Command palette, zero-leak filtering, faceted search, deep-linking.
CAP-10	Dynamic Access Control & Live Security	Container inheritance, single-doc overrides, JIT access requests, live demotion.
CAP-11	Deprecation, Trash & Enterprise Audit Ledger	Deprecation routing, 30-day recoverable trash, immutable audit log.

Dynamic Access Control & Permissions

Resolves the "binary access dilemma": rigid folders vs. over-provisioned access.

The Dynamic Access Control (DAC) Approach

- 1. **Container Inheritance:** knowledge inherits baseline access down the structure — Organization → Department → Project → Document → Sub-Document.
- 2. **Explicit Granular Overrides:** Co-owners grant explicit access to individuals or teams without altering parent boundaries.
- 3. **Internal Link Sharing:** Co-owners can enable org-wide link sharing at a configured baseline role (view or comment).

Precedence Hierarchy

Individual Explicit Grant > Team Group Grant > Project Grant > Department Baseline > Organization Default

- **Individual priority:** a direct grant to a user always overrides group or container defaults.
- **Group synergy:** membership in multiple assigned groups yields the highest privilege among them.
- **Inherited fallback:** absent an explicit grant, access flows down from Project → Department.

Six Access Roles per Document

Co-Owner · Reviewer · Editor · Commenter · Viewer · None — see the full capability grid in Section 5. Sharing is transparent: the UI always states the *source* of a user's access, e.g. "Dana — Viewer (*Inherited from Engineering Department*)" or "Charlie — Commenter (*Member of @legal-counsel*)".

Live Collaboration Security

Instant Session Transition If a role is demoted mid-session (e.g. Editor → Commenter), the live editor switches to read/comment-only via WebSocket push — no reload required.	Instant Revocation If access is fully removed, the active session terminates immediately and the user is redirected with a polite notice.
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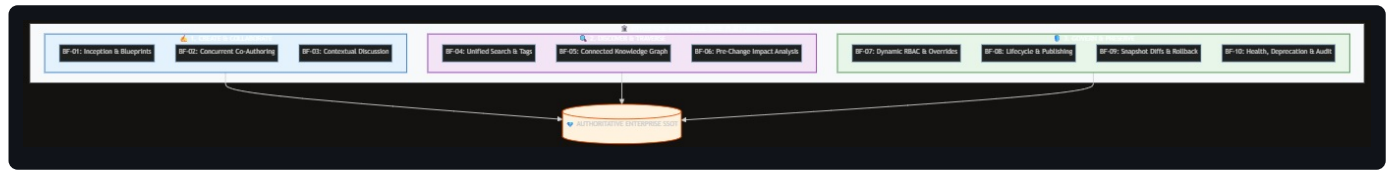
Representative Scenarios

Scenario	Access Model
Cross-functional specification	Engineering owns the doc; a Lead PM is added directly as Editor ; the Legal team is added as Reviewers — without exposing unrelated Engineering repositories.
Sensitive / confidential planning	Access is restricted to explicitly invited individuals and leadership groups; department inheritance is bypassed, keeping the document invisible to search and graph discovery for non-members.

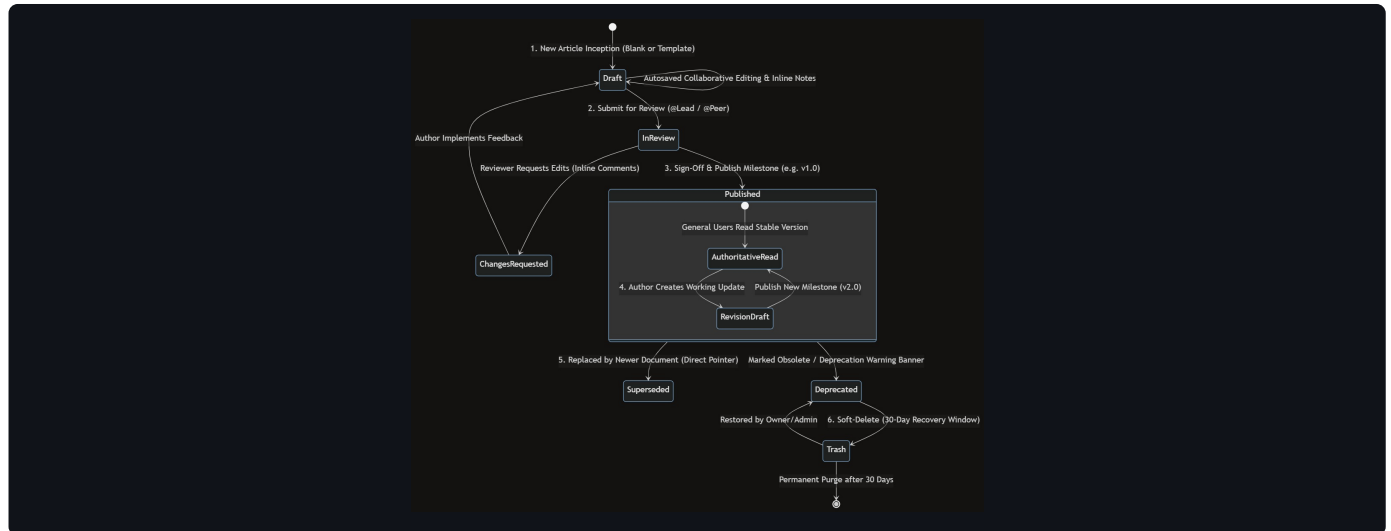
Business Workflows — Overview & Lifecycle

Ten end-to-end business flows (BF-01 – BF-10), grouped under the three operating pillars, plus the single deterministic document lifecycle every page follows.

Three Pillars → Ten Business Flows



End-to-End Master Lifecycle State Machine



Mandatory Requirement	Primary Flow	Capability Delivered
Create Articles	BF-01, BF-02	Blueprint or blank inception, rich Markdown, Mermaid diagrams, checklists.
Categories & Tags	BF-01, BF-04	Structural folders + multi-select searchable hashtags.
Search	BF-04	Command palette, instant full-text search, zero-leak RBAC filtering.
Edit / Version Articles	BF-02, BF-09	Live autosave, named milestones, visual diff, 1-click rollback.
User Comments	BF-03	Inline anchored threads, @mentions, resolve/reopen lifecycle.
Published / Draft Status	BF-08	Multi-stage lifecycle with parallel working drafts.

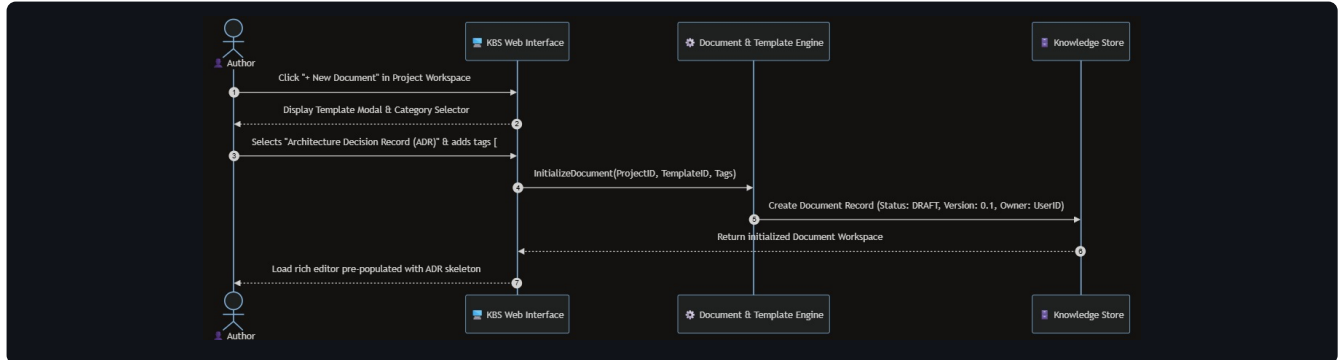
BF-01 & BF-06

Inception with categorization, and pre-change impact analysis before edits go out.

BF-01 Document Inception, Categorization & Blueprint Templates

CREATE & COLLABORATE

Ensures organizational knowledge is structured and categorized from the moment of inception using standardized blueprints (PRD, ADR, SOP, RCA) or a clean custom format.

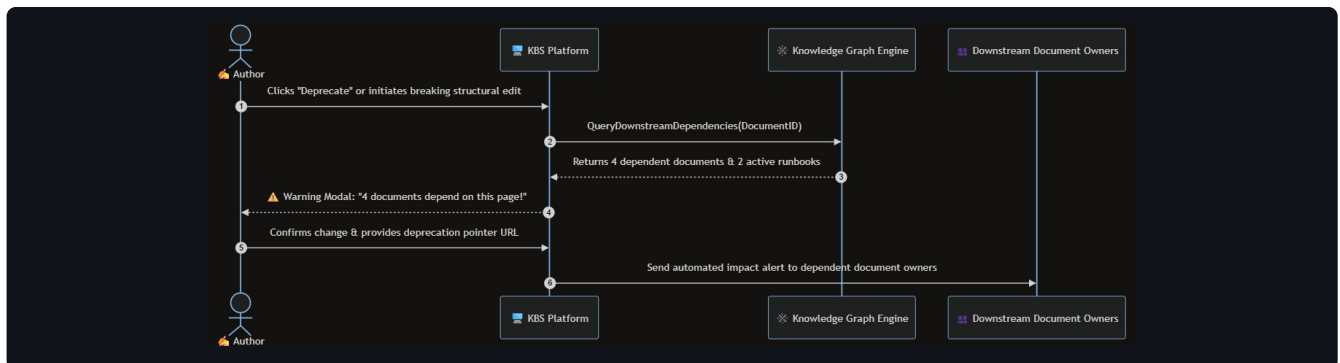


- Document automatically inherits the parent Project's default access permissions
- Author selects a standard blueprint or starts from a blank canvas
- Searchable hashtags (#auth, #database, #q3-release) added at creation
- System sets state to **DRAFT**, assigns the user as Owner, provisions a real-time room

BF-06 Pre-Change Knowledge Impact Analysis

DISCOVER & TRAVERSE

Prevents breaking downstream operational workflows by analyzing dependent documents before an author modifies, deprecates, or moves a core document.



- Author initiates deprecation or a breaking structural edit
- Knowledge Graph Engine returns all dependent documents and active runbooks
- Warning modal shows the dependency count before the change proceeds
- System sends an automated impact alert to affected downstream owners

BF-02

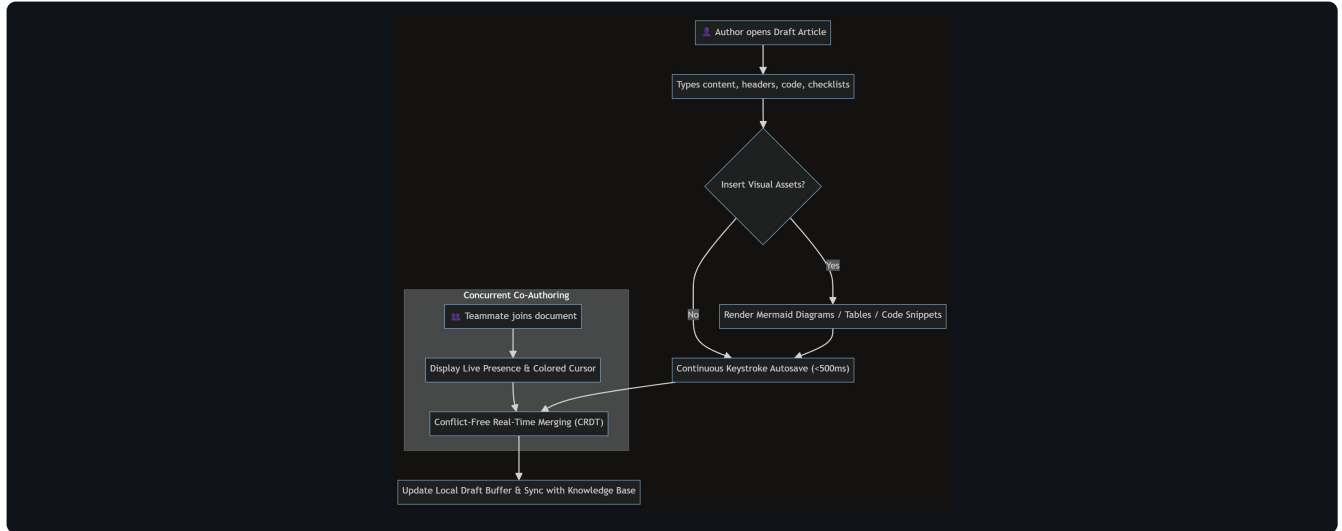
Real-Time Concurrent Authoring & Rich Content Formatting.

BF-02

Real-Time Concurrent Authoring & Rich Content Formatting

CREATE & COLLABORATE

Provides a frictionless, rich Markdown authoring environment supporting concurrent multi-user editing with zero lockouts or save conflicts.



Rich Markdown

Headings, callout banners, task checklists, tables, syntax-highlighted code blocks.

Native Diagramming

Embedded Mermaid diagrams (flowcharts, sequence, ERDs, state machines) rendered live.

Zero Save Conflicts

CRDT-based conflict-free real-time merging; continuous autosave under 500ms.

BF-03 & BF-05

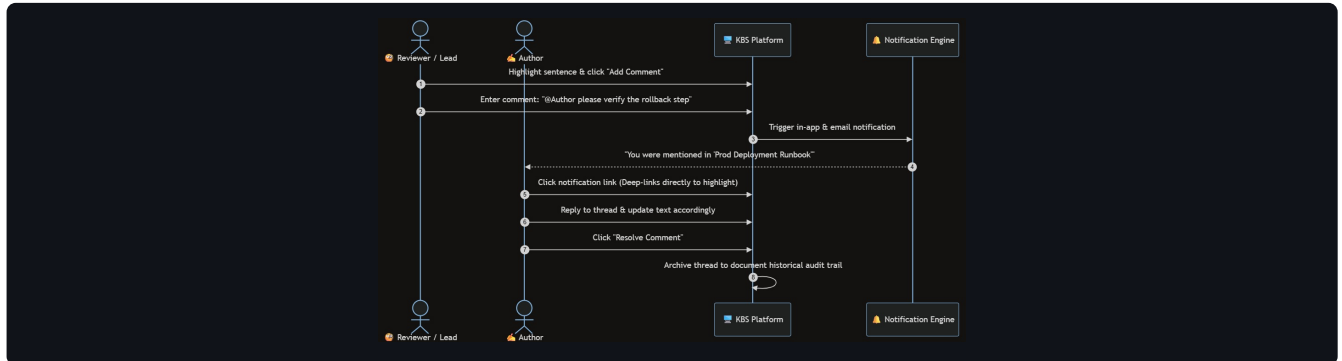
Contextual review workflow, and the connected knowledge graph that makes dependencies visible.

BF-03

Contextual Review, Inline Comments & Thread Resolution

CREATE & COLLABORATE

Transforms feedback from unstructured chat into an auditable, in-context review workflow anchored directly to relevant sentences or code blocks.



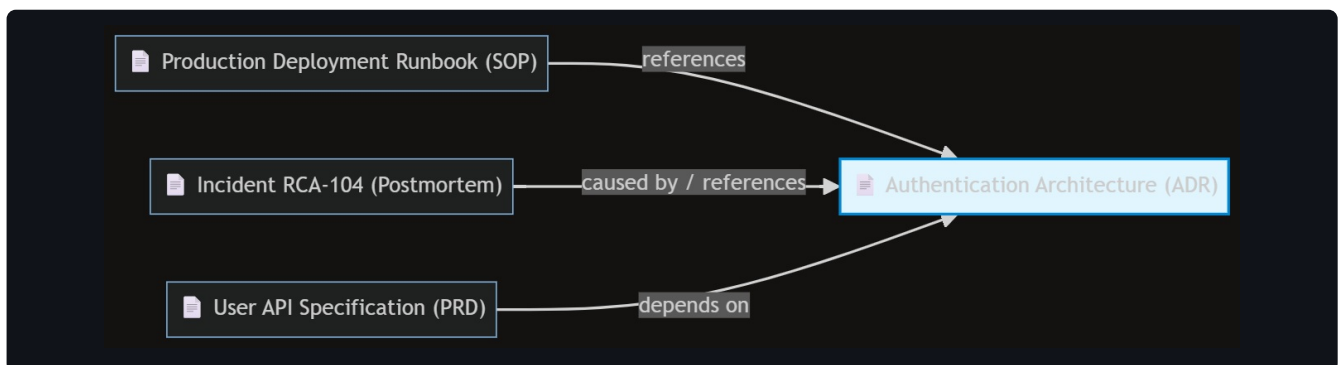
Comment lifecycle: Open / Active Resolved Reopened — resolved threads collapse from view but stay preserved in comment history.

BF-05

Connected Knowledge, Backlinks & Topology Graph

DISCOVER & TRAVERSE

Prevents knowledge isolation by automatically detecting and maintaining bidirectional relationships between documents.



- Linking Document B to Document A auto-populates A's "Incoming Backlinks" panel
- Interactive local subgraph explorer with 1-hop and 2-hop filters

BF-04

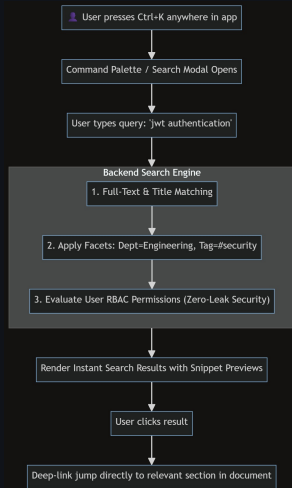
Knowledge Discovery, Semantic Search & Taxonomy Filtering.

BF-04

Knowledge Discovery, Semantic Search & Taxonomy Filtering

DISCOVER & TRAVERSE

Enables team members to instantly locate authoritative answers within seconds using an omni-present Command Palette (Ctrl+K / Cmd+K) with zero-leak security filtering.



Search Capabilities

- **Zero-Leak Security:** documents the user cannot view never appear in results, suggestions, or graph nodes
- **Faceted Filtering:** Department, Project, Author, Status, Tag
- **Command Shortcuts:** jump to recent docs, create new, switch projects
- **Deep-Linking:** clicking a result scrolls directly to the matching paragraph with a pulse highlight

BF-07

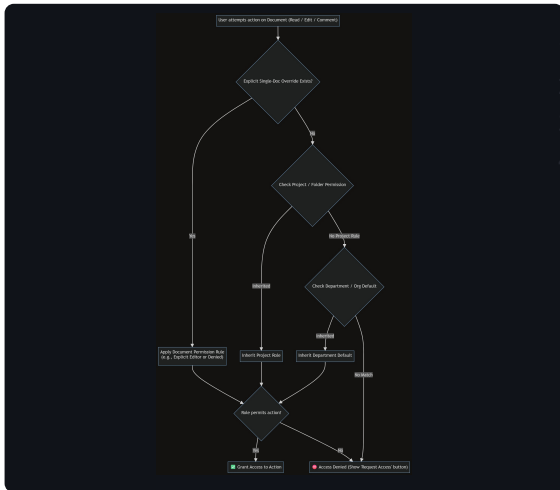
Granular Access Control & Single-Document Overrides.

BF-07

Granular Access Control & Single-Document Overrides

GOVERN & PRESERVE

Solves the "Binary Access Dilemma" by enabling hierarchical permission inheritance with explicit single-document overrides, allowing secure cross-team collaboration without compromising container security.



Evaluation Order

1. Explicit single-document override exists? → apply document rule
2. Else, check Project / Folder permission → inherit if present
3. Else, check Department / Org default → inherit if present
4. No match → Access Denied, "Request Access" shown

BF-08 & BF-09

Publishing lifecycle with reviewer sign-off, and non-destructive version comparison & rollback.

BF-08 Knowledge Lifecycle, Sign-Off & Milestone Publishing

GOVERN & PRESERVE

Ensures readers always access a verified, authoritative published milestone while authors iterate safely on a working draft.

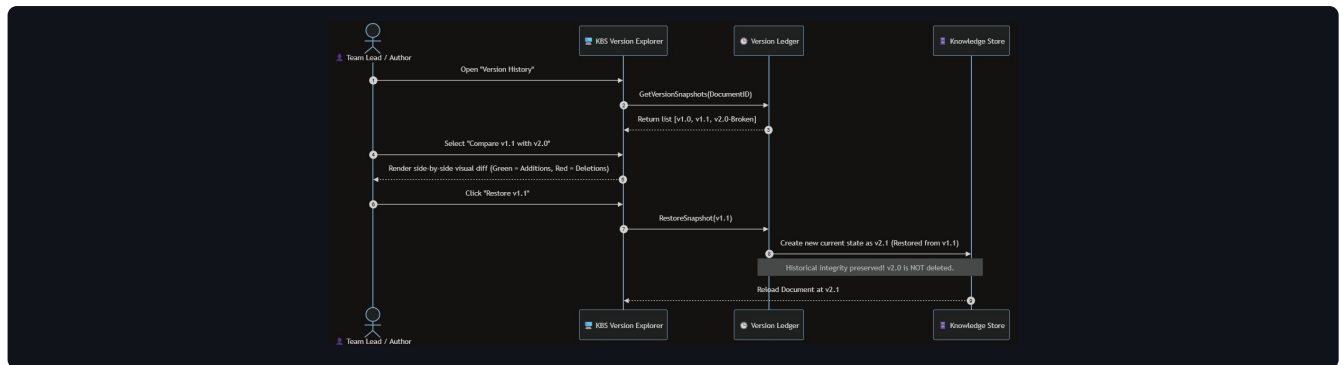


Publishing modes: Agile Direct Publish Formal Sign-Off Review — a revision draft never modifies the public version until explicitly promoted to a new milestone.

BF-09 Snapshot Versioning, Visual Diff & Safe Rollback

GOVERN & PRESERVE

Provides an immutable history of all published states with side-by-side color-coded diffing and non-destructive 1-click rollback.



Restoring a snapshot creates a new current version — historical integrity is preserved and prior milestones are never deleted.

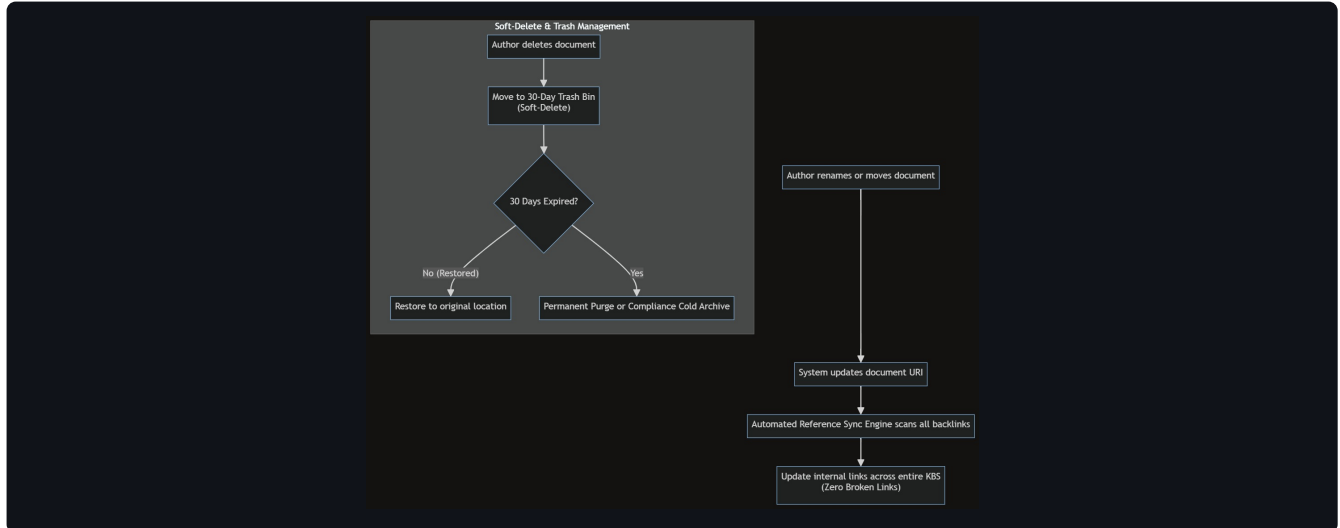
BF-10

Knowledge Health, Deprecation & Enterprise Audit.

BF-10 Knowledge Health, Deprecation & Enterprise Audit

GOVERN & PRESERVE

Prevents dead links when documents move or are renamed, and maintains an immutable audit trail of all access and content changes for compliance.



Rename / Move Resilience

Automated reference sync scans all backlinks and updates internal links platform-wide — zero broken links.

Trash & Audit

30-day recoverable trash bin before permanent purge; immutable append-only audit ledger for compliance.

Implementation Roadmap

Three phases sequence the ten business flows from MVP foundation to full enterprise governance.

Phase 1 — Core Foundation (MVP) - Document schema & Markdown editor (BF-01, BF-02) - Hierarchical categories & tags engine - Draft vs. Published state engine (BF-08)	Phase 2 — Collaboration & Search - Global search & Ctrl+K palette (BF-04) - Inline comments & @mentions (BF-03) - Milestone snapshots & visual diff (BF-09)	Phase 3 — Enterprise Governance - RBAC & single-doc override engine (BF-07) - Knowledge graph & backlink discovery (BF-05) - Impact warnings & audit ledger (BF-06, BF-10)
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Non-Functional / Operational Requirements

Dimension	Requirement
Collaboration Latency	Peer edits appear near-instantaneously (sub-second under normal network conditions).
Authorization Evaluation	Permission checks complete instantaneously without blocking page loads.
Graph & Traversal Performance	Smooth rendering of graphs and backlink queries across tens of thousands of documents.
Data Durability	Zero data loss during simultaneous edits, network drops, or tab closures.
Developer Sync Reliability	Code snippet sync and PR alerts update within seconds of GitHub events.
Accessibility	Responsive interface across desktop and mobile browsers.

Sign-Off & Acceptance Criteria

Success criteria to validate against before Phase 1 general availability.

Capability Area	Acceptance Criteria
Authoring & Formats	User can author Markdown, render Mermaid diagrams, and insert syntax-highlighted code blocks.
Categorization	Documents are organized by Project hierarchy and filterable by multi-select hashtags.
Search Performance	Ctrl+K search responds in under 200ms, returning only documents the user is authorized to read.
Version History	Users can view side-by-side visual diffs between any two versions and perform 1-click non-destructive restore.
Inline Discussions	Users can highlight text, @mention teammates, receive real-time notifications, and resolve threads.
Publishing Flow	Readers see only Published milestones while authors maintain isolated Draft revisions.
Access Control	Individual and group overrides correctly take precedence over container inheritance; live demotion applies without reload.
Audit & Compliance	All lifecycle, permission and content events are captured in an immutable, searchable audit ledger.

This document serves as the final, consolidated Business Requirements specification for the Enterprise Knowledge Base Platform (KBS), superseding the individual Part 1-5 drafts it was assembled from.